

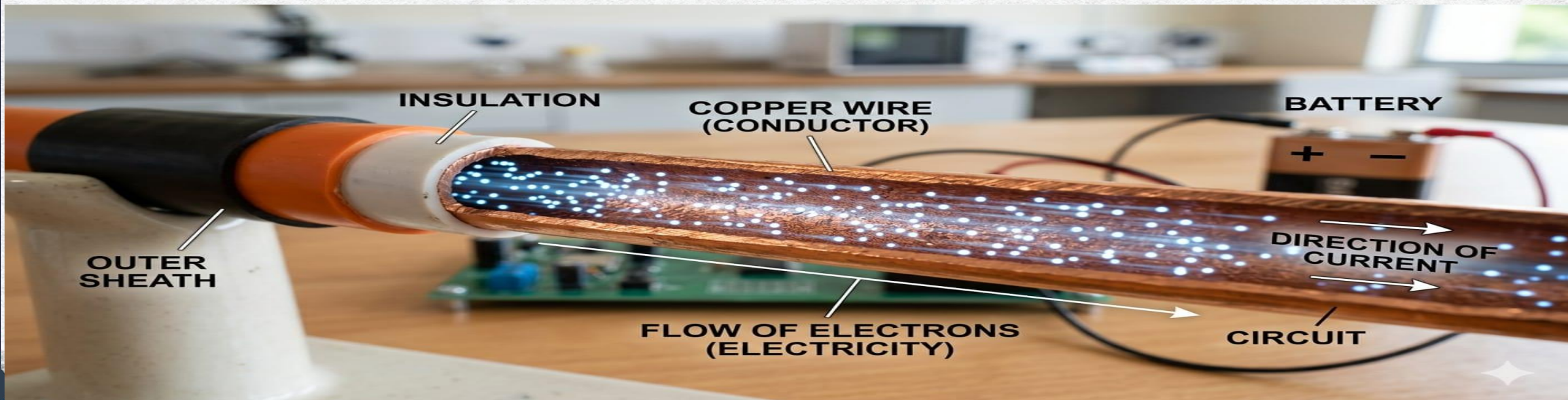


Let's Learn Basic Electronics components

Lesson-2

What is Electricity?

- ❑ Electricity is the flow of tiny particles called **electrons**.
- ❑ It helps devices like fans, lights, and mobiles to work.
- ❑ Example: When you switch on a bulb, electricity flows and it glows.

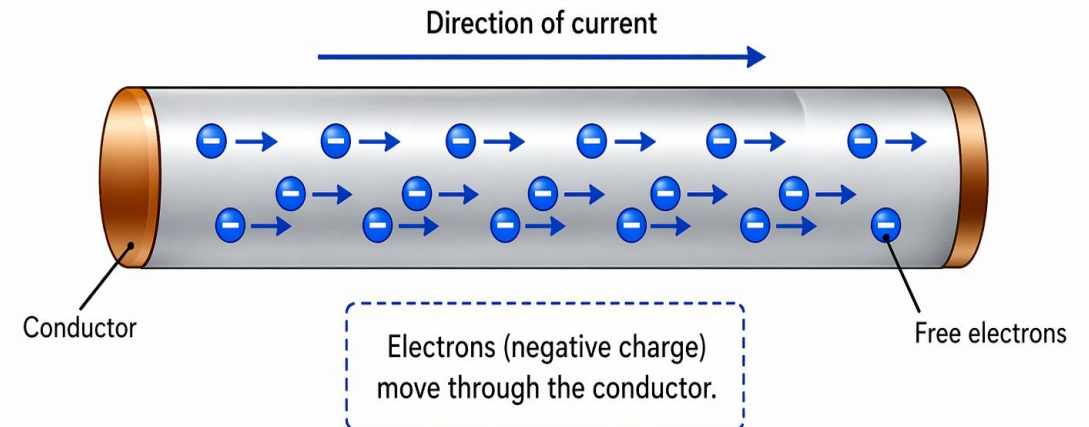


What is Electric Current ?

- ❑ Electric current is the **flow of electric charge**
- ❑ It moves through wires
- ❑ Measured in **Ampere (A)**
- ❑ Example: Like water flowing in a pipe

ELECTRIC CURRENT

Electric current is the flow of **electric charge** through a conductor.



When electrons move in an orderly way through a conductor, **electric current** is produced.

What is Conductor?

- ❑ A conductor is a material that **allows electric current to flow easily**
- ❑ It has **free electrons** that move easily

❑ Examples:

- Copper
- Aluminium
- Iron

CONDUCTOR

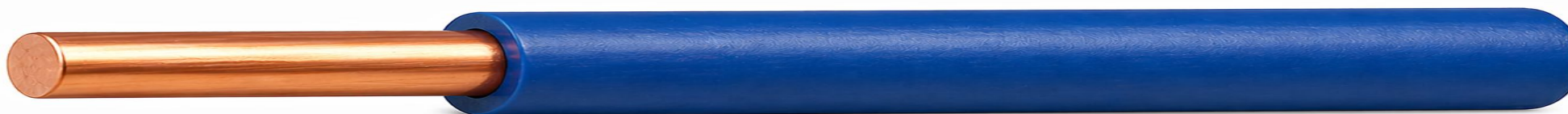


A conductor is a material that allows electric current to flow through it easily.

What is Insulator?

- An insulator is a material that **does NOT** allow electric current to flow
- Electrons cannot move freely
- **Examples: Rubber, Plastic, wood**

INSULATOR



An insulator is a material that does not allow electric current to flow through it easily.

What is Wire?

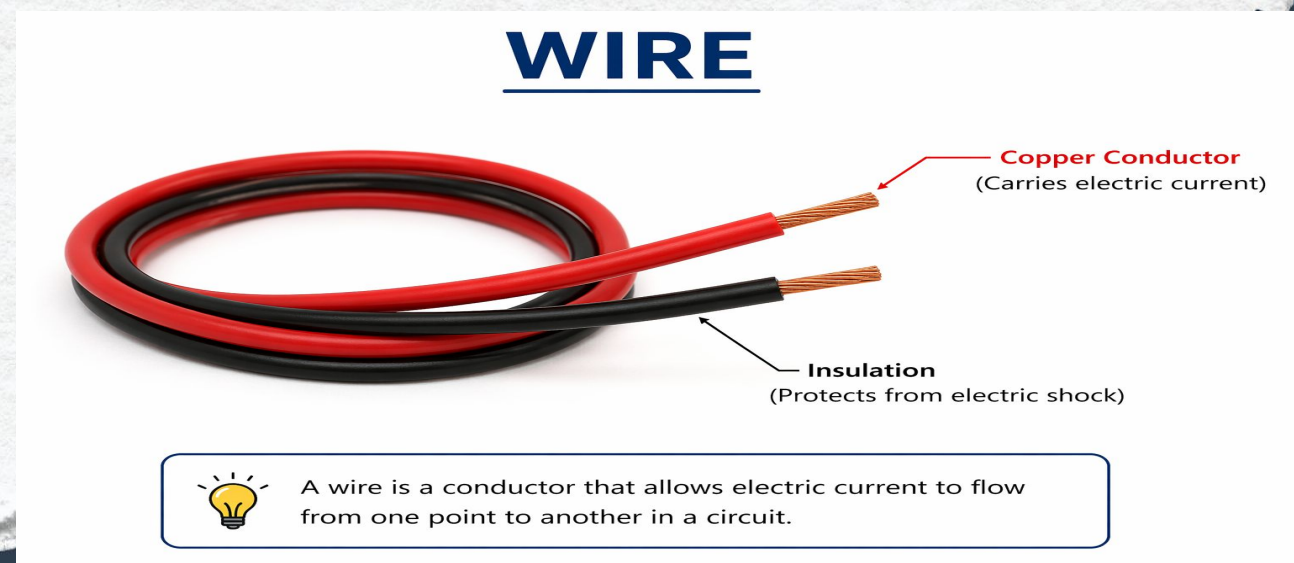
- A wire is a thin metal conductor used to carry electric current.
- It connects different components in a circuit.
- Without wires, electricity cannot flow between parts.

Conductor (Inside)

- Usually made of copper or aluminum
- Allows electricity to pass

Insulation (Outside)

- Made of plastic or rubber
- Prevents electric shock



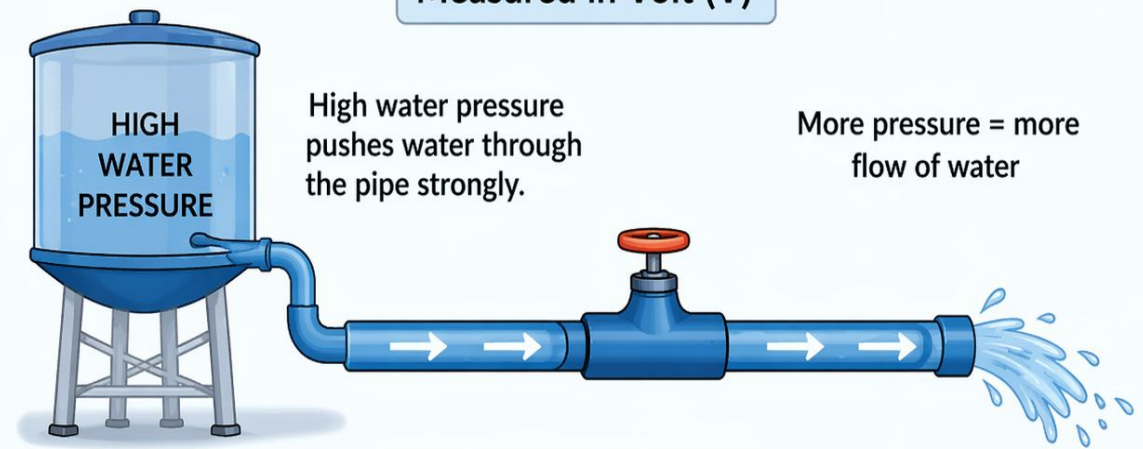
What is Voltage ?

- ❑ Voltage is the **force that pushes current**
- ❑ Measured in **Volt (V)**
- ❑ Example: Water pressure in a pipe

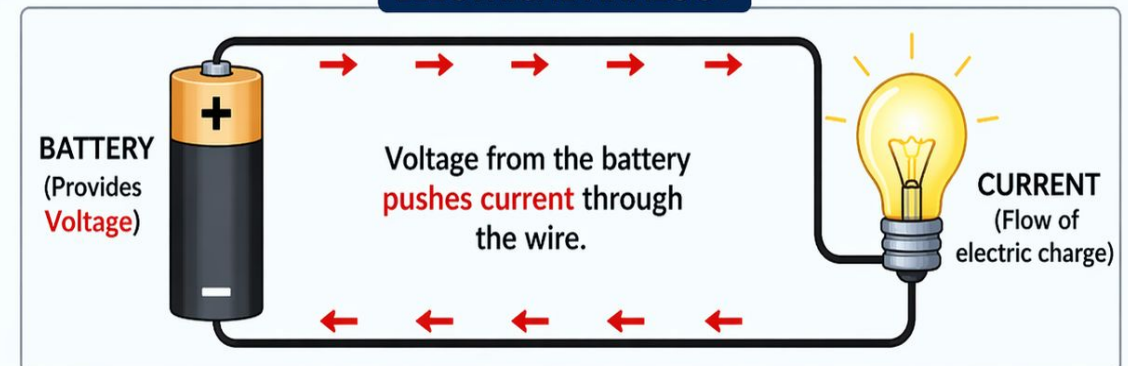
VOLTAGE

Voltage is the **force** that **pushes** current.

Measured in Volt (V)



ELECTRICAL ANALOGY



Higher **voltage** → pushes more **current** → **brighter bulb**